Michael Wieck-Sosa

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EDUCATION

Carnegie Mellon University | PhD in Statistics | Advisors: Aaditya Ramdas & Cosma Shalizi *Sept. 2022-Present*

- GPA: 3.98/4.00 | Awards: DeGroot-Goel Fellowship (2026), awarded by department faculty to one PhD student per year
- Thesis topic: high-dimensional time series | Thesis committee: Aaditya Ramdas, Cosma Shalizi, Yuting Wei, Wei Biao Wu

University of Illinois at Urbana-Champaign | MS in Statistics *May 2022*

- GPA: 3.95/4.00 | Awards: two years of assistantships with tuition waiver & stipend | GRE: 170/170 Q, 163/170 V, 4.5/6.0 W

Fordham University | BS in Mathematics with Minors in Computer Science and Economics *May 2020*

- GPA: 3.77/4.00 | Awards: *magna cum laude* | Research: deep learning-based tracking system for an endangered toad species

PROGRAMMING LANGUAGES AND TOOLS

- Python expert: extensive experience with NumPy, SciPy, pandas, scikit-learn, statsmodels, and PyTorch
- R expert: extensive experience with dplyr, Rcpp, xts, zoo, caret, mgcv, glmnet, parallel, and ggplot2
- C++ proficient: used for courses in Data Structures and Algorithms
- SQL, Git, Bash expert: extensive experience with data management, version control, and Linux

COURSEWORK

- **Statistics**: Advanced Statistical Theory, Mathematical Statistics, Time Series Analysis, Regression, Computational Statistics
- **Computer Science**: Algorithms, Data Structures, Theory of Computation, Operating Systems, Computer Architecture, Artificial Intelligence, Machine Learning, Data Mining for Listening to the Social Universe
- **Mathematics**: Stochastic Calculus, Measure-theoretic Probability, Functional Analysis, Measure Theory, Interacting Particle Systems, Geometric Flows, Differential Geometry, Lie Groupoids & Lie Algebroids, Topology, Abstract Algebra, Numerical Analysis, Numerical Linear Algebra, Real Analysis, Differential Equations, Linear Algebra, Mathematical Modeling

RESEARCH ASSISTANTSHIPS AND INTERNSHIPS

Carnegie Mellon University | NSF Grant | PI: Cosma Shalizi *May 2024-Present*

- Developed theory, methods, and software for fitting complex scientific models using non-convex optimization in Python
- Trained deep neural networks for learning generative models of multi-dimensional sequences in PyTorch using GPUs

National Center for Supercomputing Applications | Innovative Software and Data Analysis Group *Sept. 2020-May 2022*

- Built confidence bands for chemical concentrations & fluxes in water over time at 1,000+ sites using parallel computing in R

MIT Lincoln Lab | Group 38 *May 2021-July 2021*

- Tracked objects in outer space and conducted simulation studies to compare different tracking methods using MATLAB

University of Illinois at Urbana-Champaign | FORWARD Data Lab | Computer Science Department *Jan. 2021-May 2021*

- Modeled dynamics of misinformation spreading across multiple social media platforms using Hawkes processes with Python

TEACHING ASSISTANTSHIPS

- MS in Computational Finance program at CMU: Deep Learning, Natural Language Processing, Simulation Methods for Option Pricing, Financial Time Series Analysis, Financial Data Science I & II, Machine Learning Capstone Project (x2)
- BS in Statistics/Machine Learning and MS in Data Science programs at CMU: Time Series, Advanced Data Analysis

PROFESSIONAL SERVICE

- Paper reviewer | Journal of the Royal Statistical Society, Series B (Statistical Methodology) *2026*
- Committee chair | CMU Statistics PhD student committee for organizing career events *2024-2025*

SELECTED TALKS AND POSTERS

- Invited talk on deep Granger causality with transformers | CFE-CMStatistics 2026 | HTW Berlin *2026*
- Invited talk on conditional independence testing | IWSM 2026 | American University *2026*
- Contributed poster on conditional independence testing | NBER-NSF Time Series Conference | UPenn *2024*